📊 Percentile Analysis

A **percentile** shows how a value compares with the rest of the dataset. It helps us understand the **distribution and spread of data**.

Key Percentiles

Percentile	Meaning
Q1 (25%)	Bottom 25% of students
Q2 (50%)	Median (middle value)
Q3 (75%)	Top 25% of students
99%	Almost highest values
Min/Max	Lowest and highest values

SSC Percentage (ssc_p)

- **Min:** 40.89
- **Q1:** 54.5
- **Median:** 67.0
- **Q3:** 75.7
- **99%:** 87.0
- **Max:** 89.4

Explanation:

- The lowest score is **40.89**, and the highest is **89.4**
- 25% of students scored between **40.89 and 54.5**
- 50% of students scored up to **67**
- 75% scored up to **75.7**

Spread:

- Q2 – Q1 = **12.5** → moderate increase
- Q3 – Q2 = **8.7** → slightly smaller spread
- 99% to Max difference is small → very few extreme values

Insight:

- Most students scored between **54% and 75%**
- Performance is **moderately spread**

- Indicates **average and consistent performance**

HSC Percentage (hsc_p)

- **Min:** 37.0
- **Q1:** 60.6
- **Median:** 65.0
- **Q3:** 73.0
- **99%:** 91.86
- **Max:** 97.7

Explanation:

- Scores range from **37 to 97.7**
- Bottom 25% scored below **60.6**
- Half of the students scored below **65**

Spread:

- Q2 – Q1 = **4.4** → small gap
- Q3 – Q2 = **8** → larger spread in upper range
- Large gap between 99% and Max → presence of **top performers**

Insight:

- Majority scored between **60%–73%**
- Presence of **very high achievers**
- Slight variation in performance

Degree Percentage (degree_p)

- **Min:** 50.0
- **Q1:** 60.9
- **Median:** 66.0
- **Q3:** 72.0
- **99%:** 83.86
- **Max:** 91.0

Explanation:

- Scores range from **50 to 91**
- Most students fall between **60.9 and 72**

Spread:

- Q2 – Q1 = **5.1**
- Q3 – Q2 = **6**
- Gradual increase → **balanced distribution**

Insight:

- Consistent with school performance
- Indicates **stable academic trend**

MBA Percentage (mba_p)

- **Min:** 51.21
- **Q1:** 57.94
- **Median:** 62.0
- **Q3:** 66.25
- **99%:** 76.11
- **Max:** 77.89

Explanation:

- Scores range from **51.21 to 77.89**
- Majority of students scored between **57.94 and 66.25**

Spread:

- Q2 – Q1 = **4.06**
- Q3 – Q2 = **4.25**
- Very small differences → **low variability**

Insight:

- Scores are slightly lower than earlier academics
- Indicates **moderate and consistent performance**

Employability Test (etest_p)

- **Min:** 50.0
- **Q1:** 60.0
- **Median:** 71.0
- **Q3:** 83.5
- **99%:** 97.0
- **Max:** 98.0

Explanation:

- Scores range from **50 to 98**
- Bottom 25% scored below **60**
- Top students scored extremely high (**97–98**)

Spread:

- Q2 – Q1 = **11**
- Q3 – Q2 = **12.5**
- Large spread → **high variation**

Insight:

- Wide range indicates **unequal skill levels**
- Significant gap between low and high performers

Key Observation:

There is a **large skill gap among students**

🧠 **Final Overall Insights**

✅ **1. Academic Scores**

- Mostly concentrated between **60–75%**
- Show **consistent and stable performance**

⚠️ **2. MBA Scores**

- Slightly lower but **less variation**
- Indicates **uniform performance**

🚨 **3. Employability Skills**

- Highly spread out
- Shows **major differences in student abilities**

Final Conclusion

👉 *Most students have similar academic scores,*
👉 *but their skill levels (etest) vary widely,*
👉 *which can strongly impact their career outcomes.*